Bioinformatics · Week 11

Multi-Omics Paper Analysis

Analysis workflows, computational interpretation and cross-omics integration in three multi-omic studies

---

Paper 1 Spatiotemporal proteomics of human adipogenesis · Nature Metabolism, 2024

Paper 2 A universal gut-microbiome-derived signature predicts cirrhosis · Cell Metabolism, 2020

Paper 3 Shared and disease-specific host gene–microbiome associations · Nature Microbiology, 2022

Three of the four assigned papers are analysed; Paper 4 (α-ketoglutarate signalling to chromatin) was not selected. Each answer states the analytical method, the figure or table that reports it, and the key quantitative result. Where the question's premise does not match what the paper actually did, the discrepancy is flagged explicitly rather than silently accommodated.

Paper 1 Spatiotemporal Proteomics of Human Adipogenesis

Klingelhuber F., Frendo-Cumbo S., et al. Nature Metabolism 2024; 6:861–879 · Proteomics + Transcriptomics + Subcellular Spatial Proteomics

Q1. Analysis Workflow — how proteomic data were generated, quality-controlled, quantified and analysed across adipocyte differentiation

Experimental system

Four human adipocyte precursor models derived from the stromal-vascular fraction of white adipose tissue were differentiated in parallel: **SGBS** (neonatal), **primary hAPC** and **TERT-immortalised hAPC** (deliberately from the same 16-year-old male donor, so that immortalisation is the only variable between them), and **hWA** (48-year-old female) [Fig. 1a; Methods, p.874]. Differentiation was sampled at seven time points (D0, 2, 4, 6, 10, 12, 14) with $n = 3$ biological replicates each, plus a separate early series (0, 2, 12 h) to resolve the first 24–48 h [Fig. 1a legend; Extended Data Fig. 1c,d]. As an *in vivo* reference, primary adipocytes (pAC), SVF and intact WAT were profiled from seven donors (5 F/2 M; 60.7 ± 3.5 y; BMI 28.7 ± 6.6) obtained by needle aspiration and collagenase digestion [Methods, p.874]. Differentiation efficiency was verified by BODIPY imaging and marker qPCR (PLIN1, LIPE, CEBPA, PPARG).

One design decision matters for interpretation: samples were normalised by **equal total protein**, not equal cell number, and this was validated orthogonally with the **proteomic ruler** (histone intensity converted to copies per cell), because mature adipocytes accumulate a large lipid volume relative to their protein content [p.864; Extended Data Fig. 3a–c].

Acquisition: label-free single-shot DIA

- **Instrument:** Orbitrap Exploris 480 with FAIMS (compensation voltage 50) coupled to an EASY-nLC 1200; in-house 50 cm × 75 µm capillary packed with 1.9 µm C18, 60 °C; 5→45 % B over 45 min at 300 nL/min (1 h total); 500 ng loaded [Methods, p.875].
- **DIA parameters:** MS1 300–1650 m/z, 45 ms, AGC 300 %, R = 120,000; MS2 across 66 variable windows, 22 ms, AGC 1000 %, HCD 30 %, R = 15,000. DIA was chosen over DDA for higher completeness, wider dynamic range and fewer missing values [p.862; p.875].

- **Sample preparation:** 2 % SDC/100 mM Tris-HCl pH 8.5 lysis at 95 °C with sonication, BCA assay, trypsin + LysC at 1:50 overnight, TCEP/CAA reduction–alkylation, SDB-RPS StageTip desalting. Label-free quantification — no TMT, iTRAQ or SILAC anywhere in the study [Methods, p.875].

Search, quantification and quality control

Spectra were searched in **Spectronaut 15.7 directDIA** (library-free) against UniProt human canonical plus isoform entries, trypsin, 7–52 aa, two missed cleavages, fixed carbamidomethylation, variable methionine oxidation and N-terminal acetylation, 3–6 best fragments per peptide, with **precursor and protein q ≤ 1 %** [Methods, p.875–876]. Downstream analysis used Perseus 1.6.15.

- **Depth:** 5,979–7,061 protein groups per cell model, 3,638–6,403 in primary cells, **8,268 in the combined dataset**; 86 % supported by unique peptides only; five orders of magnitude dynamic range [p.862; Fig. 1b; Extended Data Fig. 1e,f].
- **Chromatographic quality:** 98 % of peptides had 3–8 data points per peak [p.862].
- **Reproducibility:** median CV 12.3 %; 72 % of peptides CV < 20 % and 42 % CV < 10 %; mean Pearson r = 0.97 between replicates [Extended Data Fig. 1h,i].
- **Normalisation and missing values:** global median normalisation in Spectronaut to correct intensity drift, followed by $\log_2$ transformation and width adjustment in Perseus. Values were retained when present in ≥ 2 replicates of ≥ 1 condition; remaining missing values were imputed from a normal distribution (downshift 1.8, width 0.3) [Methods, p.876]. No explicit batch-correction step (e.g. ComBat) is described — defensible here because all conditions were processed in one pipeline and all comparisons are within-model, but it is an omission worth noting.

Statistics and temporal modelling

The authors did **not** use limma. Differential testing used permutation-based two-sample *t*-tests and multi-condition ANOVA with **permutation-based FDR < 0.01** (S0 = 0.1 for volcano plots) [Methods, p.876]. **38 % of the proteome** changed significantly in all four models; 3,939 proteins were significant in ≥ 3/4 models, of which **1,979** were classified as conserved — defined as positive pairwise Pearson correlation of *z*-scored temporal profiles across all model pairs at shared time points [p.862–864; Fig. 1e–g]. Raw data and search output are deposited at ProteomeXchange **PXD047412**.

Q2. Data Interpretation — PCA, temporal clustering and pathway enrichment used to define conserved adipogenic programmes

PCA defines the axes of variance

PCA was run on all differentiation time points of all four models together with the *in vivo* pAC, SVF and WAT samples, using proteins with ≥ 2 valid values [Fig. 1c legend]. **PC1 explains 24.7 % and PC2 13.8 %** of variance (21.2 % for PC1 when only cell models are included) [Fig. 1c; Extended Data Fig. 2a]. Two features of the embedding carry the biological message:

- **PC1 is time.** All four models move monotonically along PC1 and converge on the position of primary adipocytes and whole WAT — the differentiation trajectory, not the cell line, is the dominant source of variance. PC1 loadings are dominated by PLIN1, PLIN4, FABP4, ACSL1, GPD1, LIPE, ABHD5, MGLL, ACACB, ACSF2, SCD and ADIPOQ, i.e. the lipid synthesis/storage module *is* the principal axis [Fig. 1d].
- **PC2–PC4 are model identity.** PC3 (8.9 %) and PC4 (5.2 %) separate hWA from the three younger male-derived models, driven by MAOA, MAOB and ADH1B — interpreted as an age/sex effect; hWA sits at a less mature position on the trajectory [p.864; Extended Data Fig. 3d,e,h,i].

Temporal clustering

Proteins significant in ≥ 3/4 models were clustered by **supervised Euclidean hierarchical clustering of *z*-scored temporal profiles**, with the four models displayed side by side so that reproducibility is visible within each cluster [Fig. 1f–h; Methods, p.876]. Clusters were then named *early*, *intermediate*, *late* and *transient* from their enrichment, and split by direction of change. Two points of accuracy: this is not fuzzy *c*-means/Mfuzz and not a GAM-based model (neither appears in the paper), and the number of clusters is not stated algorithmically — it was chosen from the heat-map structure.

The resulting three-phase programme is the core result of the paper [Fig. 1h; p.864]: **early** — down: cell cycle, ribosome biogenesis, Wnt, proteasome, spliceosome; up: glycosaminoglycan degradation, lysosome, cytochrome P450; **intermediate** — up: fatty-acid metabolism, TCA cycle, respiratory chain, ATP synthesis, PPAR signalling, peroxisome, branched-chain amino acid degradation; down: DNA replication, cytoskeleton, adhesion; **late** — up: cholesterol and triglyceride synthesis, lipid-droplet proteins; down: spliceosome and mRNA processing.

Enrichment

Enrichment used a **one-sided Fisher's exact test (over-representation analysis, not GSEA)** against UniProtKB keywords, GO, KEGG and CORUM, with an enrichment score > 2 and BH FDR < 0.1 (relaxed to 0.15 in Fig. 5a) [Methods, p.876; figure legends]. Six conserved programmes emerge, several of which are mechanistic rather than descriptive:

- **Lipid storage:** BSCL2 rises abruptly at induction; PLIN1/PLIN4, LIPE, ABHD5 rise late; PLIN2 dips transiently before rebounding [Fig. 5d].
- **Mitochondrial oxidation:** TCA cycle, respiratory chain, ATP synthase, fatty-acid oxidation, mitochondrial translation and Fe-S cluster assembly rise synchronously at the intermediate stage [Fig. 4a].
- **BCAA reprogramming:** BCAT1 falls while BCAT2 rises and **translocates from cytosol to mitochondria**, together with the remaining mitochondrial BCAA enzymes — BCAA carbon is redirected from nucleotide supply to acetyl-CoA [Fig. 4b,c; Extended Data Fig. 7].
- **One-carbon reprogramming:** the mitochondrial branch (SHMT2, MTHFD2, MTHFD1L) falls, the cytosolic branch rises, and ALDH1L1/L2 relocate into a cytosolic complex to supply NADPH [Fig. 4d–h; Extended Data Fig. 8].
- **ECM, adhesion, cytoskeleton and Wnt shut down** at the intermediate stage, while the proteasome is already low in undifferentiated SGBS [Fig. 1h].
- **Translation machinery remodelling** via NATC subunits (NAA30/NAA35) that co-sediment with the translational machinery only in mature cells [Fig. 3c,d].

> **Interpretive value of the negative result.** 1,500 proteins that never change are enriched for transcription factors, chromatin modifiers and plasma-membrane transporters [Extended Data Fig. 2p]. The adipogenic programme is therefore executed by a pre-existing transcriptional apparatus rather than by large-scale remodelling of it — a conclusion that only a proteome-wide, time-resolved design can support.

Q3. Multi-Omics Integration — transcriptome with spatial proteomics, and the evidence base for C19orf12

> **Premise correction.** The "spatial" dimension in this paper is **subcellular**, resolved by protein correlation profiling (PCP) on density gradients; it is not tissue-level spatial omics. No laser-capture microdissection, imaging mass cytometry, MIBI or CODEX was used, and there is no adipocyte-versus-immune-region comparison in tissue sections. Tissue-level context is instead provided by bulk

> proteomes of pAC, SVF and whole WAT. Reading the paper as "spatial proteomics of adipose tissue" would misattribute its main methodological contribution.

The spatial axis: protein correlation profiling

Cells were mechanically lysed and fractionated on a continuous **20–55 % sucrose gradient** (11 mL, 100,000 *g*, SW40 Ti, 3 h, 4 °C); 0.5 mL fractions were collected from the top (the first 1 mL containing the lipid-droplet layer) and each fraction was analysed by 1 h DIA [Methods, p.875; Fig. 2a]. The resulting abundance profiles across fractions were used to train a **support vector machine** (RBF kernel, sigma 0.2, $C = 8$) that assigns one- or two-compartment localisations. Coverage was 3,500–5,600 proteins per fraction; 5,675 (precursor) and 5,530 (mature) localisations were predicted, of which 4,912 and 4,452 were high-confidence, at 88 % and 96 % accuracy respectively [p.866; Extended Data Fig. 5c]. The key result is that **1,323 proteins — 654 at high confidence, roughly 20 % of the localised proteome — change localisation during differentiation**, with mitochondria, ER, endosome and lipid droplets gaining and cytosol and nucleus losing share, so that the mature localisation profile approaches that of primary adipocytes [Fig. 3a,b].

Transcriptome integration is correlative, not factor-based

The transcriptome came from the **same hAPC differentiation time course**, published previously (DDBJ DRA000991 and related accessions) [p.864; Methods, p.876]. Integration was limited to shared time points and consisted of three analyses: per-time-point **Spearman correlation** between $\log_2$ LFQ and normalised $\log_2$ counts, **Pearson correlation of *z*-scored temporal profiles**, and **1D annotation enrichment** of the correlation values; a fourth "integration" is arithmetic — localisation probabilities multiplied by temporal abundance to estimate each organelle's share of the proteome over time [Fig. 3b; Extended Data Fig. 6d].

Concordance is moderate and structured: $r = 0.39–0.48$, rising at later time points, with **16 % of proteins anti-correlated** with their transcript and protein changes generally **lagging** mRNA [Extended Data Fig. 4a–d]. Metabolic pathways are highly concordant, whereas signalling, chromatin and transcriptional regulation, and membrane-bound complexes are discordant — pointing to post-transcriptional control through degradation, translation efficiency and isoform switching (e.g. SLC25A10) [p.864, p.866]. **No WGCNA, consensus clustering, MOFA or composite integration score was used**; the paper's integration claim rests on the correlative plus localisation axes, which is a thinner integration than the question's wording implies.

How C19orf12 was established

- **Unbiased discovery.** The intersection of lipid-droplet proteomes from SGBS and hAPC gave 59 proteins, 34 of them adipocyte-specific and 29 regulated in the core temporal set; C19orf12 was selected because it had a consistent trajectory across all four models *and* the largest combined mRNA and protein induction [Fig. 5c, 6a; Extended Data Fig. 10a–c].
- **Prior knowledge.** C19orf12 is the causative gene for MPAN (NBIA4), had conflicting localisation reports (cytosol, ER, mitochondria) and a GWAS association with BMI [p.871; Extended Data Fig. 10d].
- **Localisation.** PCP gave a dual lipid-droplet plus mitochondrial profile; immunofluorescence showed the protein encircling lipid droplets and co-localising with TOM20 adjacent to droplets, identifying it as a **lipid-droplet–mitochondria contact-site protein** [Fig. 6b,c,e,f].
- **Interactome.** GFP co-immunoprecipitation with mass spectrometry (n = 4, FDR < 0.05) recovered **TOMM40, TOMM22, TOMM5** together with C12orf75, GOLGA4, PRDX1, IMMT and CANX — physically linking the droplet pool to the TOM import machinery [Fig. 6d; Extended Data Fig. 10g].
- **Loss of function phenotype.** siRNA knockdown (SMARTpool, n = 4) produced more but smaller lipid droplets, **basal lipolysis halved**, **adiponectin secretion increased ~50 %**, reduced insulin-stimulated lipid synthesis, and a **reduced maximal oxygen consumption rate** in a palmitate oxidation test that was **not rescued by extracellular palmitate** and was **abolished by etomoxir** — while maximal isoprenaline-stimulated lipolysis increased [Fig. 6g–n]. The defect is therefore specific to mitochondrial long-chain fatty-acid oxidation, and the reduced basal lipolysis is read as compensation for fatty acids that cannot be oxidised.
- **Population support.** Across six WAT transcriptome cohorts (784 individuals), C19orf12 expression correlates inversely with BMI, fat mass, adipocyte volume, triglycerides and HOMA-IR, with no sex difference; a MAGMA analysis used T2D GWAS summary statistics [Fig. 6o; Extended Data Fig. 10k].

What the evidence does not establish. There is no CRISPR knockout, no rescue experiment, no patient-derived cells or NBIA4 variant testing, and no iron or CoA biochemistry despite the iron-accumulation background of MPAN. The authors state explicitly that the molecular mechanism is unknown and offer two hypotheses: TOM40-dependent control of the mitochondrial protein complement at droplets, or a direct role in fatty-acid transfer into mitochondria [p.873]. The correct summary is that the *function* is well supported at the level of phenotype and localisation, while the *mechanism* remains unresolved.

Paper 2 A Universal Gut-Microbiome-Derived Signature Predicts Cirrhosis

Oh T.G., et al. Cell Metabolism 2020; 32:878–888 · Metagenomics + Metabolomics + Machine Learning

Q1. Analysis Workflow — processing of shotgun metagenomic and metabolomic data into disease-associated features

Cohort and phenotypes

The discovery set is the UCSD proband cohort, **163 stool samples**: 27 NAFLD-cirrhosis (16 compensated, 11 decompensated), 54 non-NAFLD controls and 82 first-degree relatives (51 non-NAFLD, 21 NAFLD without advanced fibrosis, 3 NAFLD with fibrosis, 7 NAFLD-cirrhosis) [p.2, p.8; Table S1]. NAFLD was defined as MRI-PDFF ≥ 5 % with secondary causes excluded; cirrhosis as biopsy F4, or MRE ≥ 4.69 kPa, or VCTE ≥ 11.8 kPa where MRE was contraindicated. Controls were not biopsied for ethical reasons and were classified by imaging alone [STAR Methods, p.e2].

The clinical table is essential context for every later claim, because the groups differ on almost everything confounded with cirrhosis: age 45.9 ± 19.9 vs 64.7 ± 9.8 years, BMI 26.1 vs 32.9, AST 22.3 vs 46.3 U/L, albumin 4.53 vs 4.06 g/dL, platelets 246.9 vs 149.1 $\times 10^3/\mu L$, INR 1.02 vs 1.22, type 2 diabetes 1.9 % vs 85.2 % [Table 1]. Any microbial signature will therefore be partly a marker of this clinical syndrome rather than of the microbiome per se.

Metagenomics

- **Library and sequencing:** stool and fasting serum collected on site and frozen at −80 °C; DNA with the Qiagen MagAttract PowerSoil kit (Earth Microbiome Project protocol); Nextera XT quarter reactions; 8 pM, 1 % PhiX; **Illumina MiSeq paired-end shotgun** [STAR Methods, p.e3–e4]. Read length and per-sample depth are not reported.
- **Processing chain:** adapter trimming with **skewer 0.2.2** → host depletion by **Bowtie2 against hg19** → **HUMAnN2 v0.11.2** with embedded **MetaPhlAn2 v2.7.7** for taxonomy, ChocoPhlAn pangenome alignment for known bugs, **DIAMOND against UniRef90** for unclassified reads, and mapping to **MetaCyc** for pathways → humann2_renorm_table [STAR Methods, p.e4]. Notably, no Kraken, no *de novo* assembly and no

LEfSe/MaAsLin were used.

- **Yield:** 1,216,335 gene families; 310 species across 53 families and 115 genera; 7,355 pathways; 188 strains [p.3–4].

Metabolomics

Extraction solvent was benchmarked (50 % MeOH, 80 % EtOH, water, and 2:2:1 EtOAc:MeOH:H_2O) and **50 % MeOH** selected on the basis of base-peak chromatographic complexity. Analysis used a Thermo UltiMate 3000 UPLC coupled to a **Bruker MaXis qTOF** with a 30 × 2.1 mm C18 column, 0–100 % acetonitrile (0.1 % formic acid) over 840 s at 0.5 mL/min, full scan m/z 50–2000 and top-10 CID fragmentation [STAR Methods, p.e4–e5]. Data were processed in **MS-DIAL 3.66** (MS1 tolerance 0.01 Da, MS2 0.025 Da) with annotation in **MS-FINDER 3.20**, retaining features with mass error ≤ 10 ppm and MS2 score ≥ 8/10. This produced **435 faecal metabolites**, of which 75 correlated with clinical metadata [p.4]. Polarity mode, missing-value imputation and batch correction are not described — a genuine reporting gap.

Feature selection

Rather than a single differential-abundance test, the authors combined **DESeq2 differential abundance with random-forest Gini importance** and grew or shrank the panel until the top-performing model was reached, converging on **19 species** and **17 metabolites** [p.4; STAR Methods, p.e5]. Spearman correlation linked taxa and metabolites to clinical metadata; β-diversity used weighted UniFrac with PERMANOVA ($p < 0.001$). A uniform fold-change/FDR threshold for feature inclusion is not stated. Data are at NCBI **PRJNA632613**.

Q2. Predictive Modelling — selecting, fitting and validating the signatures with random forests and ROC/AUC

Model construction

Random forests were fitted with caret::train and evaluated by **pROC**; internal performance came from **10-fold cross-validation** within the UCSD proband set [Fig. 1F, 3B legends; p.3, p.5]. External validation used three independent cohorts of increasing dissimilarity: first-degree relatives from the same centre, a Chinese Han cohort (Qin 2014, $n = 237$, quantitative metagenomics) and an Italian cohort (Iebba 2018, **16S rRNA**, genus-level). Cross-platform transfer was handled by re-standardising each feature set (preProcess plus predict) before scoring [STAR Methods, p.e5]. Key hyperparameters — ntree, mtry, node size, whether AUC was the

optimisation objective, and any handling of the 27-versus-54 class imbalance — are **not reported**, which is a reproducibility limitation.

Results

Model	Validation cohort	AUC [95 % CI]
19 species	Proband cohort, 10-fold CV	0.91 [0.89–0.93]
17 metabolites	Proband cohort, 10-fold CV	0.91 [0.89–0.93]
19 species + age	Relatives (n = 82)	0.88 [0.72–1.00]
19 species + age	China (N = 237)	0.86 [0.82–0.91]
19 species (no age)	Italy (16S)	0.89 [0.80–0.98]
19 species + age + albumin (22 features)	Proband / relatives / China	0.92 / 0.91 / 0.95 [0.92–0.98]
19 species + age + AST	Cirrhosis vs F0 / vs F1–3	0.94 [0.87–1.00] / 0.91 [0.82–1.00]

Two comparisons anchor these numbers. On the same Chinese data, the panel reached **0.86 versus 0.84** for the 15-gene marker published with that cohort [p.8]. And within a mixed-aetiology biopsy cohort, the signature separated cirrhosis from no fibrosis (0.85) and from early fibrosis (0.84) even before AST was added, rising to 0.94 and 0.91 with AST [Fig. 5E–H].

> **Three appraisal points.** (i) Feature selection was performed on the whole dataset before the 10-fold CV, with no nested CV and no held-out set, so the internal AUCs are likely optimistic. (ii) No head-to-head comparison with FIB-4, APRI, NFS or MRE is presented, which is what a clinician would ask for first. (iii) The Highlights state that microbial and metabolite signatures predict cirrhosis *independently*, yet the AUC of a combined species + metabolite model is never reported, and no DeLong test or nested-model comparison is shown — the independence claim rests on two similar single-omics AUCs plus correlation between feature sets, not on a formal test.

Q3. Multi-Omics Integration — microbial pathways, metabolites and clinical parameters into a cirrhosis signature

Pathway layer

Of 7,355 MetaCyc pathways, **39 differed** between groups, and their direction is coherent rather than random:

aromatic amino acid biosynthesis, **branched-chain amino acid (valine) biosynthesis**, chorismate biosynthesis, CoA biosynthesis II, several nucleotide biosynthetic routes (UMP, inosine-5-phosphate, guanosine and adenosine *de novo*), peptidoglycan and cell-wall synthesis, pyruvate fermentation to acetate and lactate, and starch degradation V [Fig. 2A]. The genera carrying these pathways were *Acidaminococcus* and *Veillonella* for amino acid biosynthesis, with *Clostridium*, *Bacteroides*, *Ruminococcus*, *Dialister* and *Alistipes* contributing others. Cross-cohort reproducibility of the pathway matrix was quantified with the **RV-coefficient** against the Chinese cohort and the authors' earlier cohort ($P = 6.1 \times 10^{-6}$ to 0.0052) [Fig. 2B].

Species × metabolite layer

A Spearman heat map of the 19 species against the 17 metabolites (r from −0.54 to +0.54) and a **33-node, 338-edge co-occurrence network** linking species, metabolites and discriminating gene families identified specific pairs: C18-sphingosine with *Ruminococcus gnavus*, and kynurenic acid with *Faecalibacterium prausnitzii* [Fig. 4A,B; Fig. S6A–B]. The pairings were then tested for directionality *in vitro*: anaerobic culture of *R. gnavus* ATCC 29149 across a chenodeoxycholic acid dose series (0.001–5000 µg/mL) increased C18-sphingosine and 1H-indole-3-carboxaldehyde in the supernatant [Fig. 4C; p.7]. This supplies a bile-acid → bacterium → metabolite arrow, though not causality in humans.

Clinical integration

Only three clinical variables entered the models: **age** (all models), **serum albumin** (the only metadata available in the Chinese cohort) and **AST** (to separate cirrhosis from early fibrosis). The gain is quantifiable: adding albumin moved the Chinese cohort from 0.86 to **0.95**, and adding AST moved cirrhosis-versus-F0 from 0.85 to **0.94**. BMI, ALT, bilirubin, cholesterol, GGT, glucose, insulin and platelets were examined in sensitivity analyses but not retained [Fig. S6D; p.8]. Bilirubin, INR, platelets and FIB-4 appear only in group comparisons and correlation plots, never as model features.

What makes the signature "universal"

Three lines of evidence: consistent α-diversity loss in all three countries; *Veillonella* consistently increased and *F. prausnitzii* and *Eubacterium rectale* consistently decreased across the US, Chinese and Italian studies; and monotonic behaviour of signature species along NAFL → F1–3 → F4 in a mixed-aetiology cohort (alcohol, HBV, HCV, NAFLD) [Fig. 1B, 5I–J; p.9]. The argument is explicitly that heterogeneity of aetiology, diet and ethnicity strengthens rather than weakens generalisability.

Biological reading. The paper argues that microbially encoded aromatic and branched-chain amino acid biosynthesis is dysregulated in cirrhosis, that glycochenodeoxycholic acid rises (an FXR ligand and NASH drug target), that vitamin D and its derivatives fall, and that tryptophan rises while kynurenic acid falls — a shift toward the indole route [p.4, p.7]. Worth noting as a limit of the paper's own framing: the up-signature is dominated by oral commensals (*Streptococcus parasanguinis*, *S. salivarius*, *Veillonella* spp.), which invites the oral-to-gut translocation and portosystemic-shunt interpretation, but small-intestinal bacterial overgrowth, shunting and salivary influx are **not discussed** anywhere in the text. That reading is inference, not a stated conclusion. The authors are careful to state that the associations do not demonstrate causality [p.9].

Paper 3 Shared and Disease-Specific Host Gene–Microbiome Associations

Priya S., et al. Nature Microbiology 2022; 7:780–795 · Transcriptomics + Microbiomics + Sparse Multivariate Integration

Q1. Analysis Workflow — processing and comparing paired host RNA-seq and 16S data across CRC, IBD and IBS

Design

The study assembled **208 paired host-transcriptome/16S pairs (416 samples)** from colonic mucosal biopsies, in three disease contexts with **no subjects shared between them** [Abstract; Methods, p.787]:

- **CRC:** 88 pairs = 44 patients × (tumour + matched normal tissue); 23 F/21 M, mean age 65 (17–91); resection specimens from Minnesota Bionet, snap-frozen, pathology-confirmed. The transcriptomes were newly generated for this study.
- **IBD:** 78 pairs = 34 Crohn's disease + 22 ulcerative colitis + 22 non-IBD; 38 F/40 M; from HMP2 (ibdmdb.org), biopsies in RNAlater.
- **IBS:** 42 pairs = 29 IBS + 13 healthy; 31 F/11 M, mean age 38 (20–63); Mayo Clinic; some individuals contributed two time points, which were averaged.

Host transcriptome

CRC: ~100 mg tissue in Qiazol, AllPrep DNA/RNA, RIN ≥ 8 and ≥ 1 µg input, TruSeq stranded total RNA, HiSeq 2500; FastQC, Trimmomatic, **HISAT2 → hg38** (mean alignment 87.11 %, 14.4–31.5 M aligned reads per sample), featureCounts [Methods, p.787; Extended Data Fig. 4]. **IBD:** TruSeq stranded mRNA, HiSeq 2000/2500 101 bp paired-end, Broad Picard pipeline. **IBS:** TruSeq v2, HiSeq 2000, MAP-RSeq v2 (TopHat2 → hg19).

> **Hidden batch structure.** Three cohorts, three aligners, two genome builds (hg38 vs hg19). This heterogeneity is not incidental: it is why the authors refuse to pool the data, and it constrains how much of any "shared" signal can be attributed to biology rather than to processing.

Normalisation and the deliberate absence of batch correction

Within each cohort: protein-coding genes only (biomaRt) → expressed in ≥ 50 % of samples → **DESeq2 VST** → genes below the 25th percentile of variance removed, leaving **12,513 (CRC), 11,985 (IBD) and 12,429 (IBS)** genes [Methods, p.788]. The authors explicitly reject **ComBat** and removeBatchEffect, arguing that batch-correction tools are built for case–control differential testing and would distort an integration analysis. Instead each cohort is integrated independently and combined only at the final comparison step as a meta-analysis [Methods, p.788]. Covariates were limited to sex and disease subtype (CD/UC; IBS-C/IBS-D); BMI, biopsy site, medication and diet were unavailable.

16S and taxon matrices

CRC used **V5–V6** on MiSeq 2 × 250 with USEARCH v7 merging and QIIME 1.7 OTU picking; IBD used **V4** with UPARSE; IBS used **V4** with shi7 and BURST [Methods, p.787–788]. The pipeline is OTU-based; a DADA2 comparison in a subset of CRC samples gave phylum $\rho = 0.89$, class 0.68, genus 0.60, so OTU-level resolution is a real (if bounded) limitation. After removing archaea, chloroplasts and kit/environmental contaminants, taxa were aggregated at six ranks and filtered to those with **relative abundance ≥ 0.001 in ≥ 10 % of samples**, then concatenated into a combined matrix (**235 / 121 / 238** taxa) and **CLR-transformed** [Methods, p.788]. Alpha and beta diversity were not recomputed; published results were cited instead.

Comparing across diseases

Global concordance between the two matrices was tested before any integration: **Procrustes** (Aitchison distance for host, Bray–Curtis for microbes, 9,999 permutations) gave $P = 1 \times 10^{-4}$ for CRC but 0.1 for IBD and 0.42 for IBS; Mantel tests gave 0.0026, 0.2597 and 0.9525 [Fig. 1b; Methods, p.789]. Only CRC shows whole-matrix correspondence, which is exactly the case for methods that assume a sparse, few-genes-to-few-taxa architecture. Comparability across diseases was then enforced by using one identical pipeline and tuning strategy in each cohort, and by never fitting a pooled model. Notably, **no differential expression or differential abundance analysis was performed**; "disease-specific" is defined by modelling cases and controls separately and taking the case-only signal [Methods, p.790–791].

Q2. Integration Algorithms — sparse CCA, LASSO and pathway enrichment

Matrices

X = CLR-transformed taxa (235/121/238); **Y** = VST gene expression (12,513/11,985/12,429); both standardised, and each cohort split into case and control models [Methods, p.788–789]. This split is the methodological backbone: every subsequent comparison is between separately fitted models, not between groups within one model.

Sparse CCA (pathway/group level)

- **Tool and objective:**PMA v1.1 in R 3.3.3, maximising $\mathbf{u}^T\mathbf{X}^T\mathbf{Y}\mathbf{v}$ subject to L1 constraints $\|u\|_1 \leq \lambda_1$, $\|v\|_1 \leq \lambda_2$ [Methods, p.789].
- **Tuning:** PMA's built-in permutation search is only one-dimensional, so the authors wrote a **grid search with leave-one-out cross-validation**, selecting the penalty with the highest CV correlation: **CRC 0.15 (taxa)/0.20 (genes), IBD 0.177/0.333, IBS 0.4/0.1** [Methods, p.789].
- **Significance:** no permutation test. Instead, for each of the first ten components, the held-out score $\mathbf{X}_i\mathbf{u}_{-i}$ **versus** $\mathbf{Y}_i\mathbf{v}_{-i}$ was correlated across samples and tested, with BH FDR < 0.1. This tests a **component** (a gene set against a taxon set), not individual features [Methods, p.789].
- **Result:7 (CRC), 4 (IBD) and 6 (IBS)** significant components, averaging 828 genes + 8 taxa, 2,095 + 6, and 577 + 61 respectively. There is no hard loading threshold — the top ten features per side are shown, with node size scaled by |coefficient|; explained covariance is not reported [Methods, p.789–790].
- **Enrichment:** Fisher's exact test with the CCA input genes as background, MSigDB canonical **KEGG + PID** sets, excluding sets with < 25 or > 300 genes or < 5 overlapping genes, BH FDR < 0.1. Redundancy among the top 15 pathways was collapsed using an **overlap coefficient of 0.5** into connected components, keeping the largest representative per component [Methods, p.789–790; Fig. 2a].

LASSO (single gene–single taxon level)

This is the step most often misread: LASSO was used for **sparse inference, not prediction**. One model was fitted **per host gene**, using ~200 taxon abundances as predictors (glmnet 2.0-13, λ by LOOCV), with sex and disease subtype forced into the design and then filtered out [Methods, p.790]. Inference used **desparsified (debiased) LASSO** via hdi to obtain confidence intervals and *p*-values, followed by BH FDR and intersection

with **stability selection** (stabs, 100 subsamples of *n*/2, selection frequency threshold 0.6). There is no AUC, no ROC curve and no held-out test set, because prediction was not the objective. Over 12,300 gene-wise models per cohort (about 5.5 CPU-hours each) yielded **755 (CRC), 1,295 (IBD) and 441 (IBS)** significant associations, spanning 745 × 120, 1,246 × 56 and 436 × 102 gene–taxon pairs. Enrichment of these hits used KEGG + PID + Reactome, filtering 1,881 sets down to about 99 tested per cohort, giving **18 disease-unique pathways (4 CRC / 9 IBD / 5 IBS)** [p.782; Methods, p.790].

Robustness checks are unusually thorough: case and control models share **zero** significant associations (preserved after downsampling to equal sample size); the differential-enrichment test compares log odds ratios between case and control models (FDR < 0.2); and a loss-of-function analysis looking for control-specific pathways found nothing beyond housekeeping biology [Methods, p.790–791].

Q3. Biological Interpretation — shared versus disease-specific regulatory structure

Operational definitions

A **shared pathway** is significant in all three cohorts; a **disease-specific pathway** is significant in only one. A **shared microbe or gene** is significant in ≥ 2 diseases [Fig. 2a, 4a,b]. Because the cohorts were modelled independently, "shared" means **independent rediscovery**, not an artefact of pooling — this is the strongest design feature of the paper.

Core finding: pathways are shared, microbes are not

Shared pathways converge on **inflammation, barrier protection/repair and energy metabolism**: oxidative phosphorylation, RAC1 signalling, proteasome, focal adhesion, cell adhesion molecules, integrin and syndecan-2 [Fig. 2a; p.781]. The cleanest illustration is RAC1: the *same* gene set (CYBB, NCF1, CYFIP2, PLCB2, MAP3K1, STAT3, CTNNB1, IQGAP3, NCKAP1, ARPC1B, CFL1, ARHGDIA, BCAR1) is coupled to **entirely different taxa in each disease** — *Streptococcus*, Synergistales and GN02 in CRC; *Granulicatella* and *Clostridium sensu stricto 1* in IBD; *Bacteroides massiliensis*, *Bifidobacterium* and *Odoribacter* in IBS [Fig. 2b]. Correspondingly, the genes shared by all three diseases are enriched for VEGF signalling, complement and coagulation cascades and cytokine–cytokine receptor interaction, while their partner taxa are disease-specific [Fig. 2d].

Disease-specific associations

- **CRC** (102 CCA-level pathways include 52 CRC-specific; 4 LASSO-level): IL-10 signalling, NOTCH1 in cancer, regulation of MECP2. Anaerolineae and TM7 correlate negatively with CXCL8 and IL1RN, Bacteroidales positively with CCR2 and FPR1; *Parvimonas* and *Bacteroides fragilis* pair with the CRC-specific syndecan-1 pathway; *Blautia* is negatively associated with RIPK3 and ZBP1 [Fig. 2c, 3c,e; p.782–783]. Worth flagging: *Fusobacterium nucleatum* appears only in the introduction, never in the results — the oncobacterial evidence here is carried by *Parvimonas*, *B. fragilis* and *Streptococcus*.
- **IBD** (25 CCA-level, 9 LASSO-level): integrin beta-1 and A6B1/A6B4 integrins, thrombin signalling through PARs, glucagon-type ligand receptors. Peptostreptococcaceae correlates positively with MAPK3, VIPR1, PYGB and NCK2 [Fig. 2c, 3d].
- **IBS** (25 CCA-level, 5 LASSO-level): B-cell receptor signalling, ribosome, sumoylation of DNA damage response proteins, arachidonic acid metabolism and PPAR signalling. *Prevotella*–SMC5 is negative, *B. massiliensis*–PLA2G4A positive [Fig. 2c, 3e].

How the networks were built — and what they are not

Nodes are genes/pathways (circles) and taxa (triangles); edges are LASSO-significant associations; edge **weight is Spearman ρ rather than the LASSO coefficient**, because the coefficient is biased; each shared node retains at most the top ten edges by q-value; layout in Cytoscape 3.5.1 [Fig. 3 legend; Methods, p.791]. The one graph-theoretic operation is the connected-component collapse used to de-redundify pathways. There is **no community detection, no module identification and no hub metric** (degree, betweenness), so these should be described as bipartite association graphs — "regulatory network" is a figure of speech here, not a computed structure.

Shared statistics and the key interpretive control

Three taxa are shared by all three diseases (**Peptostreptococcaceae, Streptococcus, Staphylococcus**), 20 appear in ≥ 2 (unique: 104 CRC, 42 IBD, 89 IBS). Five genes are shared by all three, 135 in ≥ 2 (unique: 653 CRC, 1,124 IBD, 375 IBS) [Fig. 4a,b]. The decisive control is the overlap of the *input* matrices: **84 %** of host genes are common inputs across diseases but only **~37 %** of taxa, and disease-specific taxa appear in the other two cohorts' inputs only **12 %** of the time [Methods, p.791]. Disease specificity therefore originates on the microbial side; it is not an artefact of genes being unavailable in one cohort.

Conclusions

1. **Taxonomic similarity is not functional similarity.** Shared taxa and shared pathways coexist, but the pairing between them is disease-specific.
2. **Rewiring.** The same gene can be linked to different microbes (PINK1 with *Collinsella* in CRC, Peptostreptococcaceae in IBD, *Blautia* in IBS; TLR4 with GN02 in CRC and Acidaminococcaceae in IBD), supporting the authors' hypothesis that phenotype arises from how a microbe engages host programmes rather than from composition alone [Fig. 4b; p.786–787].
3. **Three biological axes** organise the results: inflammation/innate immunity (IL-10, TLR4, complement–coagulation, NK cytotoxicity), barrier and repair (RAC1, integrins, syndecan-1, HAS2, ANXA1) and metabolism/energy (OXPHOS, butyrate-producing *Blautia*, arachidonic acid, PPAR).

Limitations. The design is cross-sectional, so direction is unresolved — diseased mucosa may simply permit colonisation by a given taxon. Only taxonomy is measured, with no microbial functional potential. Age, biopsy site, diet, host genetics and medication are unadjusted; platforms, genome builds and OTU pipelines differ irreconcilably; control groups are small (22 in IBD, 13 in IBS); and concatenating six taxonomic ranks risks double counting. The honest reading is that this is a hypothesis-generating map of host–microbe association structure, not a causal model.

Appendix Cross-Paper Comparison

The same three analytical problems — normalisation, dimensionality, and validation — solved differently depending on data type

Dimension	Paper 1 (adipogenesis)	Paper 2 (cirrhosis)	Paper 3 (host–microbe)
Layers	Proteome (DIA), transcriptome, subcellular localisation	Metagenome, faecal metabolome, clinical metadata	Host transcriptome, 16S microbiome
Scale	4 models × 7 time points × *n* = 3; 8,268 protein groups	163 samples; 310 species, 435 metabolites; 3 validation cohorts	208 pairs / 416 samples; ~12,000 genes, 121–238 taxa
Normalisation	Equal protein loading + median normalisation; proteomic-ruler validation	HUMAnN2 renormalisation; sqrt/scale for metabolome (batch effects unaddressed)	VST for genes; CLR for taxa; **no ComBat** — meta-analysis instead
Dimensionality	PCA (PC1 = time, 24.7 %), hierarchical clustering of *z*-scored trajectories	Random-forest Gini importance for feature selection	Sparse CCA (group level) + gene-wise desparsified LASSO (feature level)
Integration logic	Correlation of temporal profiles + localisation × abundance; no latent-factor model	Parallel signatures, pathway-level convergence, clinical covariates	Independent per-cohort models, then intersection (meta-analysis)
Validation	Cross-model reproducibility (≥ 3/4 models), proteomic-ruler recount, 784-person expression cohorts	10-fold CV + three external cohorts incl. cross-platform 16S	Zero case/control overlap, downsampling, input-overlap control
Headline result	38 % of proteome remodelled; 1,979 conserved; 1,323 relocalised	AUC 0.91 single-omics, 0.95 with age + albumin in China	Pathways shared, microbes disease-specific; 3 shared taxa
Main caveat	Correlative integration; C19orf12 mechanism unresolved (siRNA only)	Feature selection before CV; no FIB-4/APRI comparison; combined model not reported	Cross-sectional; taxonomy only; heterogeneous platforms

Method reporting follows the papers' own Methods and figure legends. Where a method named in the assignment question does not appear in the paper (subcellular rather than tissue-level spatial proteomics in Paper 1; no WGCNA or factor model in Paper 1; no LEfSe or combined-omics AUC in Paper 2; no ComBat, no differential-expression step and no hub/community detection in Paper 3), the answer states what was actually done and marks the divergence, rather than substituting the assumed method.